

Aquaculture Intelligence System (AIS)

5-Week Day-by-Day Implementation Blueprint & Multi-Agent Specification

Document Title:	5-Week Master Implementation Blueprint — AIS
Target Domain:	Aquaculture / AgriTech / Multi-Model AI Engineering
Target Species:	Rohu, Catla, Tilapia, Pangasius, Shrimp
System Stack:	React 18 PWA, FastAPI, PostgreSQL 16 / TimescaleDB, pgvector, MinIO, ONNX/TFLite
AI Modules:	6 Core Modules (Water Quality, Fish Disease, Growth, Feed, Satellite, Gen AI Advisory)
Execution Policy:	User handles Model Training; Agent handles Scaffolding, APIs, UI, Exports & QA

PRIMARY EXECUTION POLICY & RESPONSIBILITY MATRIX:

- **USER RESPONSIBILITY:** Model dataset training, hyperparameter execution, and accuracy verification on local machine/GPU.
- **AGENT RESPONSIBILITY:** Code scaffolding, ETL pipelines, database migrations, REST/WebSocket API endpoints, ONNX/TFLite export pipelines, frontend PWA UI, Docker orchestration, and unit test suites.

1. System Architecture & AI Modules Summary

Module	Primary Task	Model / Algorithm	Runtime Format	Target Metric
1. Water Quality	Anomaly & 72h Forecast	Isolation Forest, XGBoost (x3)	joblib / FastAPI	DO MAPE < 8%
2. Fish Disease	15-Class & Batch Detection	EfficientNet-B2, YOLOv8-nano	ONNX / TFLite FP16	Top-1 Acc > 88%
3. Growth Forecaster	Growth & Harvest Date	Bioenergetic SGR + XGBoost	joblib / FastAPI	Yield RMSE < 12%
4. Feed Optimizer	Daily Feed & FCR Track	XGBoost Regressor + Rules	joblib / FastAPI	Feed MAPE < 8%
5. Satellite Monitor	NDWI & Pond Boundary	ResNet34 U-Net + Indices	PyTorch / ONNX	Boundary IoU > 90%
6. Gen AI Advisory (Last)	RAG Advice & PDF Report	Claude API + pgvector HNSW	REST (SSE) + PyMuPDF	Faithfulness > 91%

2. Granular 5-Week Day-by-Day Execution Plan (35 Days)

WEEK 1: Foundation, Infrastructure & Water Quality Monitor

Day	Task Description & Engineering Work	Role	Target Deliverable
Day 1 (W1D1)	Repo scaffold, .env.example, requirements.txt, Docker Compose (FastAPI, TimescaleDB, Redis, MinIO, Mosquitto).	Agent	Running docker stack with /health check.
Day 2 (W1D2)	Execute PostgreSQL/TimescaleDB migration SQL scripts for hypertables (sensor_readings, rag_chunks).	Agent	Database hypertable migration executed.
Day 3 (W1D3)	Build Mosquitto MQTT subscriber bridge in Python (pond/+/sensors) with Pydantic v2 validation.	Agent	ingestion/mqtt_bridge.py writing live data.
Day 4 (W1D4)	Build tabular preprocessing pipeline: range checks, Z-score spike filter, missing value imputation, diurnal features.	Agent	preprocessing/tabular.py feature engine.
Day 5 (W1D5)	Scaffold Isolation Forest anomaly detector & XGBoost 24/48/72h forecasting trainer scripts (train_water_quality.py).	User (Train) / Agent	Model joblib artifacts in registry.
Day 6 (W1D6)	Develop FastAPI REST endpoints (/sensors/ingest, /sensors/history, /predict/water-quality, /predict/stress-risk).	Agent	Tested Water Quality REST endpoints.
Day 7 (W1D7)	Develop React 18 PWA frontend shell, Pond Card Grid, Chart.js live sensor time-series charts, and stress risk badges.	Agent	Week 1 Staging MVP Deployed.

WEEK 2: Deep Learning — Fish Disease & Stress Detector

Day	Task Description & Engineering Work	Role	Target Deliverable
Day 8 (W2D1)	Configure MinIO S3 bucket & OpenCV image quality filter (resolution >= 64px, Laplacian blur score > 80).	Agent	preprocessing/image.py ETL pipeline.
Day 9 (W2D2)	Build Albumentations image augmentation pipeline & Claude API rare-class synthetic image generator script.	Agent	Image loader & dataset train/val/test splits.
Day 10 (W2D3)	Scaffold PyTorch EfficientNet-B2 classifier backbone (progressive unfreezing) & training script train_disease_efficientnet.py.	User (Train) / Agent	EfficientNet-B2 weight checkpoint.
Day 11 (W2D4)	Build Ultralytics YOLOv8-nano multi-fish batch detection script for net-pull photos & trainer train_yolov8.py.	User (Train) / Agent	YOLOv8 weights file (yolov8n_fish.pt).
Day 12 (W2D5)	Build model export pipeline: PyTorch -> ONNX (cloud serving) & TFLite FP16 (<15MB for offline Android).	Agent	Exported ONNX and TFLite artifacts.
Day 13 (W2D6)	Construct ICAR-CIFA Disease Knowledge JSON store & FastAPI endpoints /predict/disease and /predict/batch-health.	Agent	Disease REST endpoints functional.
Day 14 (W2D7)	Develop React PWA Camera Widget component, diagnosis summary modal, and IndexedDB offline sync handler.	Agent	Week 2 Staging MVP Deployed.

WEEK 3: Bioenergetics — Fish Growth & Feed Optimization Engine

Day	Task Description & Engineering Work	Role	Target Deliverable
Day 15 (W3D1)	Build bioenergetic SGR calibration engine with ICAR-CIFA species lookup tables and biomass equation.	Agent	modules/growth/bioenergetic s.py.
Day 16 (W3D2)	Scaffold XGBoost growth forecaster & stocking recommender models script train_growth_models.py.	User (Train) / Agent	Growth model joblib artifacts.
Day 17 (W3D3)	Develop binary search harvest date predictor & Logistic Regression mortality risk trainer train_mortality_model.py.	User (Train) / Agent	Harvest date & mortality scoring engine.
Day 18 (W3D4)	Build Feed Optimization Engine (temperature/DO rate adjustments) & XGBoost feed regressor trainer train_feed_optimizer.py.	User (Train) / Agent	Trained feed regressor module.
Day 19 (W3D5)	Implement dawn/dusk feed timing scheduler, FCR drift tracker (FCR > 1.15x target alert), and feed budget calculator.	Agent	Feed optimization business logic.
Day 20 (W3D6)	Implement 6 REST endpoints: /stocking/record, /predict/growth, /predict/harvest-date, /optimize/feed, /track/fcr.	Agent	Functional Growth & Feed API catalogue.
Day 21 (W3D7)	Develop React UI components: Growth Curve Chart, Harvest Date Countdown, Daily Feed card, FCR alert banner.	Agent	Week 3 Staging MVP Deployed.

WEEK 4: Remote Sensing (Drone/Satellite Pond Monitor)

Day	Task Description & Engineering Work	Role	Target Deliverable
Day 22 (W4D1)	Implement Celery task for weekly Sentinel Hub API Sentinel-2 L2A GeoTIFF download (bands B02-B11) with cloud filter (<20%).	Agent	modules/satellite/ingest.py pipeline.
Day 23 (W4D2)	Develop NDWI, Chlorophyll-a proxy, Turbidity index computation algorithms and 30-day baseline change rasters.	Agent	Satellite index computation module.
Day 24 (W4D3)	Implement ResNet34 U-Net segmentation architecture for pond boundary extraction (>90% IoU) & train_unet_segmentation.py.	User (Train) / Agent	U-Net segmentation weight checkpoint.
Day 25 (W4D4)	Configure GeoTIFF tile server in MinIO S3 object storage and cache 7-day satellite rasters.	Agent	MinIO satellite tile cache.
Day 26 (W4D5)	Build surface anomaly classification handler for farmer-uploaded drone photos.	Agent	Drone photo classifier handler.
Day 27 (W4D6)	Implement FastAPI satellite endpoints (/satellite/pond-map/{farm_id}, /satellite/bloom-risk/{farm_id}, /drone/analyze).	Agent	Satellite REST API endpoints.
Day 28 (W4D7)	Develop React Leaflet.js interactive map component rendering color-coded NDWI/Chlorophyll overlays with timeline controls.	Agent	Week 4 Staging MVP Deployed.

WEEK 5: Generative AI Advisory Engine (Going Last) & Production Launch

Day	Task Description & Engineering Work	Role	Target Deliverable
Day 29 (W5D1)	Build PyMuPDF text extractor & RecursiveCharacterTextSplitter for ICAR-CIFA/FAO PDFs (~2,000 pages).	Agent	etl/rag_chunker.py text chunks.
Day 30 (W5D2)	Configure pgvector extension (HNSW index) & embedding generator script generate_embeddings.py (all-MiniLM-L6-v2).	User (Train) / Agent	pgvector indexed PostgreSQL store.
Day 31 (W5D3)	Develop Claude API orchestration engine (modules/genai/orchestrator.py) with function calling tools.	Agent	LLM Agent router with tool calls.
Day 32 (W5D4)	Integrate Whisper ASR voice transcription, Google TTS / Bhashini speech output, and ReportLab PDF report generator.	Agent	Voice query & ReportLab PDF service.
Day 33 (W5D5)	Develop React PWA Chatbot UI (streaming SSE text, voice mic, citations, PDF download button).	Agent	Generative AI Chatbot PWA UI.
Day 34 (W5D6)	Perform cross-module microservice integration across all 6 modules, shared TimescaleDB/Redis, and Claude router.	Agent	Fully integrated microservices stack.
Day 35 (W5D7)	Deploy production stack via Docker Compose on Cloud VM with Nginx SSL, Sentry monitoring, and MLflow snapshot.	Agent	Live Production AIS MVP Deployed.